

CHAPTER THREE – SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

This chapter presents the analysis and design of Campuz. It outlines the functional and non-functional requirements, system architecture, user roles, use case analysis, database design, and user interface design. The purpose of this chapter is to provide a blueprint for the implementation of the proposed academic communication platform.

3.2 Existing System Analysis

Currently, academic communication within universities relies heavily on general-purpose messaging platforms such as WhatsApp and Telegram, as well as institutional learning management systems such as KNUST V-Class.

While these systems provide useful communication and content-sharing capabilities, they suffer from several limitations including poor organization of academic information, lack of intelligent reminders, difficulty retrieving important resources, and inadequate mechanisms for ensuring critical updates reach students.

These limitations motivate the development of Campuz as a specialized academic communication platform.

3.3 Proposed System

Campuz is a mobile-based academic communication platform designed to facilitate structured communication between students, course representatives, and lecturers.

The platform organizes academic information into Academic Communication Hubs where announcements, resources, assignments, schedules, and discussions can be managed efficiently.

The system also incorporates smart reminders, AI-assisted information retrieval, and optional SMS redundancy for urgent academic updates.

3.4 User Roles

The system consists of three primary user roles:

Student

A student can:

- Create an account.
- Join academic hubs using invitation links or QR codes.
- View announcements.
- Access shared resources.
- Search academic information.
- Receive reminders and notifications.
- Interact with the AI Academic Assistant.

Hub Administrator (Course Representative)

A course representative can:

- Create academic hubs.
- Generate invitation links and QR codes.
- Post announcements.
- Upload lecture slides and resources.
- Share meeting links.
- Mark announcements as urgent.
- Manage hub content.

Lecturer (Optional Future Extension)

A lecturer may:

- Upload learning materials.
- Post official announcements.
- Collaborate with course representatives.
- Share academic resources.
- Monitor hub activities.

3.5 Functional Requirements

The system shall:

- Allow user registration and authentication.
- Allow users to join hubs using QR codes or invitation links.

- Allow administrators to create and manage hubs.
- Allow administrators to post announcements.
- Allow administrators to upload academic resources.
- Provide search functionality.
- Generate deadline reminders.
- Support AI-assisted information retrieval.
- Deliver notifications for important updates.
- Support optional SMS fallback for urgent announcements.

3.6 Non-Functional Requirements

The system shall:

- Be easy to use.
- Provide a responsive user interface.
- Support Android mobile devices.
- Ensure data security and privacy.
- Provide reliable information delivery.
- Maintain acceptable performance under normal usage conditions.
- Support future scalability and feature expansion.

3.7 System Architecture

Campuz adopts a client-server architecture.

Components include:

1. Mobile Application (Flutter)
2. Backend API Service
3. Database
4. Notification Service
5. AI Assistant Module

The mobile application communicates with the backend through REST APIs. The backend manages user authentication, data storage, notifications, and communication between system components.

3.8 Use Case Analysis

Major use cases include:

Student

- Register Account
- Login
- Join Hub
- View Announcements
- Download Resources
- Search Information
- Receive Notifications
- Use AI Assistant

Course Representative

- Create Hub
- Share Hub Link
- Generate QR Code
- Post Announcement
- Upload Resource
- Mark Announcement as Urgent

3.9 Database Design

The proposed database consists of the following entities:

Users

- user_id
- full_name
- phone_number
- email
- role
- profile_picture

Hubs

- hub_id
- hub_name
- description
- created_by
- join_code

Deadlines

- deadline_id
- hub_id
- title
- due_date
- description
- created_at

Memberships

- membership_id
- user_id
- hub_id

Announcements

- announcement_id
- hub_id
- title
- content
- is_urgent
- created_at

Resources

- resource_id
- hub_id
- title
- file_url
- file_type
- uploaded_by

- uploaded_at

Notifications

- notification_id
- user_id
- title
- content
- is_read

3.10 User Interface Design

The user interface consists of:

- Splash Screen
- Onboarding Screens
- Registration Screen
- OTP Verification Screen
- Profile Setup Screen
- Home Screen
- Academic Hub Screen
- Announcement Screen
- Resource Screen
- AI Assistant Screen
- Profile Screen

The interface is designed to be simple, intuitive, and familiar to users accustomed to modern messaging applications.

3.11 System Components

Authentication Module

Responsible for:

- registration
- login

- OTP verification
- profile setup

Academic Hub Module

Responsible for:

- creating hubs
- joining hubs
- managing memberships

Announcement Module

Responsible for:

- posting announcements
- urgent notices
- notifications

Resource Management Module

Responsible for:

- slides
- PDFs
- links
- academic materials

AI Assistant Module

Responsible for:

- searching information
- retrieving announcements
- answering academic queries

Reminder Module

Responsible for:

- deadline tracking
- notification scheduling
- calendar synchronization

3.12 Technology Stack

Component	Technology
Mobile App	Flutter
Backend API	Django REST Framework
Database	PostgreSQL
Authentication	Firebase Auth /Custom Auth
Notifications	Firebase Cloud Messaging
Version Control	Git & GitHub
AI Module	OpenAI API (Future Integration)

3.13 Assumptions and Constraints

Assumptions

- Users possess smartphones.
- Internet access is available.
- Users have basic mobile app literacy.

Constraints

- SMS delivery depends on third-party providers.
- AI responses depend on internet connectivity.
- Initial version targets Android devices.

3.14 Summary

This chapter presented the analysis and design of Campuz, including system requirements, user roles, architecture, database structure, interface design, system components, technology stack, and

assumptions and constraints. These designs serve as the foundation for the implementation phase described in subsequent chapters.